

MVP (Phase 1)

User Authentication & Account Management

- Authentication Service
 - Email/password authentication
 - JWT access/refresh token management
 - Role-based access control (Consumer, Business, Captain, NGO, Admin)
 - User Profile Management
 - Personal information management
 - Language preference (Arabic / English)
 - Address management
 - Notification preferences
-

Consumer Marketplace

- Deal Discovery Engine
 - Nearby discounted products listing
 - Geolocation-based search
 - Product filtering
 - Category
 - Distance
 - Discount percentage
 - Pickup/Delivery availability
- Product Listing Module
 - Product images
 - Original vs discounted pricing
 - Expiration countdown
 - Available quantity display
 - Store information

- Search Service
 - Full-text product search
 - Category-based search
 - Store-based search
-

Business Portal

- Business Registration & Verification
 - Store onboarding workflow
 - Store profile management
 - Location management
 - Inventory Publishing
 - Manual product creation
 - Bulk inventory upload
 - Product image management
 - Expiration date assignment
 - Quantity management
 - Listing Management
 - Publish/unpublish listings
 - Inventory status monitoring
 - Real-time stock updates
-

AI Product Recognition

- Image Processing Pipeline
 - Product image upload
 - OCR extraction
 - Expiration date detection
- Product Classification

- Product type identification
 - Metadata extraction
 - Confidence scoring
 - Validation Layer
 - Product verification workflow
 - Human correction fallback
-

Dynamic Pricing Engine

- Rule-Based Pricing
 - Expiration-based discount calculation
 - Configurable discount thresholds
 - Minimum/maximum discount limits
 - Automated Price Updates
 - Scheduled repricing jobs
 - Real-time price recalculation
 - Price history tracking
-

Ordering System

- Cart Management
 - Add/remove items
 - Quantity management
 - Price validation
- Checkout Flow
 - Order creation
 - Pickup selection
 - Delivery selection
- Order Lifecycle

- Pending
- Confirmed
- Preparing
- Delivered
- Cancelled

Delivery Management

- Delivery Assignment Service
 - Nearest captain matching
 - Availability tracking
 - Assignment automation
- Delivery Tracking
 - Order status updates
 - ETA calculations
 - Delivery completion confirmation

Trust & Safety

- Food Verification
 - AI verification indicators
 - Store verification indicators
 - Safety status display
- Store Reputation System
 - Customer ratings
 - Review collection
 - Trust score calculation

Administration Portal

- User Management
 - Consumer management
 - Store management
 - Captain management
 - Listing Moderation
 - Product review
 - Abuse investigation
 - Analytics Dashboard
 - Active users
 - Orders
 - Revenue recovery metrics
-

V2 (Growth & Optimization)

Personalized Recommendation Engine

- Recommendation Service
 - User behavior tracking
 - Purchase history analysis
 - Preference modeling
 - Personalized Feed
 - Recommended products
 - Recommended stores
 - Dynamic ranking algorithm
-

Reservation & Availability Management

- Inventory Reservation System
 - Temporary stock locking
 - Reservation timeout management

- Concurrency handling
-

Smart Notification System

- Event-Driven Notifications
 - New nearby deals
 - Price drops
 - Expiring reservations
 - Favorite store updates
 - Notification Channels
 - Push notifications
 - Email notifications
-

Charity Integration Platform

- NGO Management
 - NGO onboarding
 - Eligibility verification
 - Donation Marketplace
 - Surplus allocation workflows
 - Donation claim process
 - Pickup scheduling
 - Impact Reporting
 - Donation reports
 - Food recovery metrics
-

Business Analytics Suite

- Revenue Recovery Dashboard
 - Recovered revenue metrics

- Unsold inventory trends
 - Product performance analysis
 - Listing Performance Metrics
 - Conversion rates
 - Sell-through rates
 - Demand analysis
-

Customer Support Platform

- Ticket Management
 - Issue creation
 - Escalation workflows
 - Resolution tracking
 - Dispute Resolution
 - Order dispute handling
 - Refund management
-

V3 (Intelligence & Scale)

Predictive Demand Forecasting

- Demand Prediction Models
 - Historical demand analysis
 - Seasonal forecasting
 - Location-based demand forecasting
 - Inventory Risk Detection
 - Overstock prediction
 - Waste probability scoring
-

AI-Driven Discount Optimization

- Dynamic Pricing Models
 - Demand-aware pricing
 - Conversion probability prediction
 - Revenue optimization algorithms
 - Pricing Simulations
 - Expected sell-through estimation
 - Revenue impact forecasting
-

Automated Inventory Monitoring

- Inventory Monitoring Agent
 - Expiration risk detection
 - Unsold stock identification
 - Automated intervention triggers
 - Business Recommendations
 - Discount recommendations
 - Donation recommendations
-

Compliance & Governance

- Food Safety Compliance
 - Product eligibility validation
 - Donation eligibility validation
 - Compliance auditing
 - Audit & Monitoring
 - Activity logging
 - Pricing decision history
 - Inventory change history
-

Advanced Business Intelligence

- **Forecasting Dashboard**
 - Demand forecasts
 - Inventory forecasts
 - Revenue projections
 - **Operational Insights**
 - Waste reduction trends
 - Product lifecycle analytics
 - Store performance benchmarking
-

Platform Infrastructure Optimization

- **Performance & Scalability**
 - Caching strategies
 - Background job processing
 - Database optimization
- **Reliability & Monitoring**
 - System observability
 - Error tracking
 - Automated alerting
 - Backup & recovery systems